JUBAYER MAHMUD NIRJUS

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EDUCATION

B.Sc. in Veterinary Science and Animal Husbandry **Nov 2023 – Dec 2028 (Expected)**
Patuakhali Science and Technology University (PSTU), Bangladesh
Academic Status: Undergraduate Student
Research Focus: AI for Animal Welfare, Precision Livestock Farming, One Health & AMR.

RESEARCH PROFILE

Undergraduate researcher in Veterinary Science and Animal Husbandry working at the intersection of artificial intelligence, animal welfare, precision livestock farming, and One Health. Research experience spans PRISMA-ScR-compliant scoping reviews, computer vision for animal welfare, multi-district epidemiological field studies, antimicrobial resistance (AMR), and farm-level data analytics.

RESEARCH INTERESTS

- **AI for Animal Welfare:** Computer Vision, Behaviour Recognition, Automated Welfare Assessment.
- **Precision Livestock Farming (PLF):** Wearable Sensors, Edge AI, Time-Series Analysis & Signal Processing.
- **One Health & AMR:** Antimicrobial Use Patterns, Epidemiological Risk Determinants, Ethnoveterinary Practices.

PUBLICATIONS

- **Nirjus, J. M.**, Saha, M., & Sarker, R. (2026). Ethnoveterinary herbal practices are associated with reduced within-farm antibiotic use and improved farm performance in commercial poultry in Bangladesh: A multi-district cross-sectional study. *Journal of Ethnopharmacology*, 373, 122316. DOI: 10.1016/j.jep.2026.122316
- **Nirjus, J. M.**, & Whittaker, A. L. (2026). Computer vision for welfare monitoring in farmed ruminants: A scoping review of evidence, framework alignment, and deployment readiness. *animal*, 102002. DOI: 10.1016/j.animal.2026.102002
- **Nirjus, J. M.**, Hasan, M. S., Islam, M., Saif, M. S. B., Pulok, M. Z. I., Shahriar, A., Jerin, R., Hoque, S., Mojumder, P., Asha, S. N., Islam, T., Unnoby, S., Das, S., Hossain, M. R., Khan, M. R. B. R., Saha, M., & Thomson, P. (2026). Antimicrobial use patterns, risk determinants, and farmer typologies in commercial poultry production in Bangladesh: A multidimensional cross-sectional analysis under the One Health framework. *Preventive Veterinary Medicine*, 258, 107057. DOI: 10.1016/j.prevetmed.2026.107057
- **Nirjus, J. M.** (2025). Cultural drivers of antimicrobial misuse: The impact of religious festivals on antibiotic use in ruminants in Bangladesh. *Journal of Research in Veterinary Sciences*, 5(3), 302–316. DOI: 10.5455/jrvs.20250711071818

HONORS, AWARDS & RESEARCH GRANTS

- **Winner & Project Grant Recipient – SEAOHUN Regional Student Competition** **2026**
  Southeast Asia One Health University Network (SEAOHUN) | Funded by Chevron & FAO (UN)
  - Selected among 9 winning international teams across Southeast Asia, India, and Bangladesh in the *One Health Campus-to-Community (OH-C2C) Challenge*; awarded project implementation funding of up to USD 2,000.

INTERNATIONAL CONFERENCE PRESENTATIONS

- **Oral Presentation:** "Intelligent Compassion: The Role of Artificial Intelligence in Transforming Animal Welfare Across Veterinary, Agricultural, and Conservation Contexts." *5th International Paris Congress on Agriculture & Animal Husbandry*, Paris, France (June 2026).
- **Oral Presentation:** "Beyond the Prescription — A One Health Approach to AMR." *10th International Halich Congress on Multidisciplinary Scientific Research*, Istanbul, Turkey (May 2026).

RESEARCH EXPERIENCE

Student Research Collaborator — AI, Animal Welfare & Computer Vision **2025 – Present**

Collaborative Research with Associate Professor Alexandra L. Whittaker (University of Adelaide, Australia)

- Contributed to a PRISMA-ScR-compliant scoping review evaluating 3,129 records across 11 databases on computer vision in farmed ruminants; led evidence synthesis and manuscript drafting; paper published at *animal — The International Journal of Animal Biosciences*.
- Synthesizing sensor mechanics, TinyML/Edge-AI feasibility, and signal processing architectures across 46 studies for an ongoing review manuscript on wearable sensor analytics.

Student Research Collaborator — One Health, AMR & Epidemiology **2025 – Present**

Collaborative Research with Assistant Professor Mousumi Saha (Gazipur Agricultural University - GAU)

- Co-led two multi-district cross-sectional field studies investigating ethnoveterinary practices and antimicrobial use patterns in commercial poultry; lead papers published in *Journal of Ethnopharmacology* and *Preventive Veterinary Medicine*.
- Assisting in data processing, epidemiological modeling, and literature synthesis for clinical studies on zoonotic risk factors and bovine health.

Co-Investigator — OneHealthVillage Project **2026 – Present**

Grant-Funded Project | SEAOHUN, Chevron & FAO (UN) | Supervisor: Prof. Dr. Farzana Islam Rume (PSTU)

- Supporting implementation of a community-based One Health intervention in Babuganj, Barishal, focused on AMR awareness and zoonotic risk through behavioural surveys and courtyard-based education.

Undergraduate Researcher **Dec 2023 – Present**

Faculty of Animal Science and Veterinary Medicine, PSTU

- Leading questionnaire development, multi-district cross-sectional field data collection, farmer interviews (KoboToolbox), and data cleaning targeting AMR epidemiology and livestock production performance.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL.
- **Python Ecosystem:** NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch.
- **AI & Computer Vision:** Computer Vision, Object Detection, Behaviour Recognition.
- **Signal & Sensor Analytics:** Time-Series Analysis, Signal Processing, Wearable Sensor Data Processing.
- **Data Analysis & Statistics:** SPSS, Logistic Regression, Cluster Analysis, Statistical Analysis, Microsoft Excel.
- **Research Tools & Methods:** Covidence, PRISMA-ScR, Overleaf (LaTeX), KoboToolbox.
- **Bioinformatics:** Sequence Alignment, Basic Genomic Data Analysis.

SELECTED CERTIFICATIONS & TRAINING

- **Genomics & Computational Biology:** Algorithms for DNA Sequencing, Python for Genomic Data Science (Johns Hopkins University, Coursera) **2026**
- **Animal Behaviour & Welfare:** Animal Behaviour and Welfare, Animal Welfare in the Clinic (The University of Edinburgh) **2025**
- **Data Science & AI:** Data Analysis with Python, Python for Data Science, AI & Development (IBM / Coursera) **2025**
- **Scientific Writing:** Writing in the Sciences (Stanford University) **2025**

LEADERSHIP, ADVOCACY & SERVICE

- **Associate Member (Overseas Studies Team)**, PSTU Research Society **Dec 2025 – Present**
- **Executive Member (Volunteer & Logistics)**, Bioinformatics Club, PSTU **Aug 2025 – Present**
- **Founder**, Animal Cortex (*Science Communication Platform for Animal AI & Welfare*) **Jul 2025 – Present**
- **Assistant Organizing Secretary**, VASA (Vet Sci & AH Students' Assoc.) **Jul 2026 – Present**
- **Community Outreach:** Founder, Blood Warriors Society; Assistant Campus Ambassador, WSDA NZ.